

He-Chen Weng

✉ he-chen_weng@brown.edu

🌐 inspirehep.net/authors/2172748

Research Interests

Scattering Amplitudes, Perturbative Amplitude Bootstrap, Cluster Algebra, Non-perturbative Bootstrap

Education

- PhD in Physics, *Brown University* Sept 2023 - Present
Advisor: Professor Marcus Spradlin
- BS in Physics, *National Taiwan University* Sept 2017 - Jun 2022
- BS in Electrical Engineering, *National Taiwan University* Sept 2017 - Jun 2022

Publications

1. L. Ferro, T. Lukowski, L. Ren, M. Spradlin, A. Volovich, **H.-C. Weng** and Y.-Q. Zhang, *de Sitter Wavefunction from Quadrangular Polylogarithms: Chain Graphs*, [arXiv:2605.06542 \[hep-th\]](#)
2. S. Paranjape, M. Skowronek, M. Spradlin, A. Volovich and **H.-C. Weng**, *Landau Analysis of One-Cycle Negative Geometries*, [arXiv:2604.22683 \[hep-th\]](#)
3. S. Paranjape, M. Skowronek, M. Spradlin, A. Volovich and **H.-C. Weng**, *Cluster Bootstrap for Cosmological Correlators*, JHEP **09**, 121 (2026), [doi:10.1007/JHEP09\(2026\)121](#) [arXiv:2603.08670 \[hep-th\]](#)
4. S. Paranjape, M. Spradlin, A. Volovich and **H.-C. Weng**, *Split-Helicity Tree Amplitudes and Flag Cluster Algebras*, JHEP **05**, 130 (2026), [doi:10.1007/JHEP05\(2026\)130](#), [arXiv:2512.09704 \[hep-th\]](#)
5. A. Pokraka, M. Spradlin, A. Volovich and **H.-C. Weng**, *Symbol Alphabets in QCD and Flag Cluster Algebras*, JHEP **03**, 120 (2026), [doi:10.1007/JHEP03\(2026\)120](#), [arXiv:2506.11895 \[hep-th\]](#)
6. R. Bhardwaj, M. Spradlin, A. Volovich and **H.-C. Weng**, *Unitarity of bespoke amplitudes*, Phys. Rev. D **110**, no.10, 106016 (2024), [doi:10.1103/PhysRevD.110.106016](#), [arXiv:2406.04410 \[hep-th\]](#)
7. L.-Y. Chiang, Y.-t. Huang and **H.-C. Weng**, *Bootstrapping string theory EFT*, JHEP **05**, 289 (2024), [doi:10.1007/JHEP05\(2024\)289](#), [arXiv:2310.10710 \[hep-th\]](#)
8. L.-Y. Chiang, T.-C. Huang, Y.-t. Huang, W. Li, L. Rodina and **H.-C. Weng**, *The geometry of the modular bootstrap*, JHEP **02**, 209 (2024), [doi:10.1007/JHEP02\(2024\)209](#), [arXiv:2308.11692 \[hep-th\]](#)
9. L.-Y. Chiang, Y.-t. Huang, L. Rodina and **H.-C. Weng**, *De-projecting the EFThedron*, JHEP **05**, 102 (2024), [doi:10.1007/JHEP05\(2024\)102](#), [arXiv:2204.07140 \[hep-th\]](#)
10. L.-Y. Chiang, Y.-t. Huang, W. Li, L. Rodina and **H.-C. Weng**, *(Non)-projective bounds on gravitational EFT*, [arXiv:2201.07177 \[hep-th\]](#)
11. L.-Y. Chiang, Y.-t. Huang, W. Li, L. Rodina and **H.-C. Weng**, *Into the EFThedron and UV constraints from IR consistency*, JHEP **03**, 063 (2022), [doi:10.1007/JHEP03\(2022\)063](#), [arXiv:2105.02862 \[hep-th\]](#)

Work in Progress

1. L.-Y. Chiang, Y.-t. Huang and **H.-C. Weng**, *Bootstrapping the Heterotic String*
2. L. Ferro, T. Lukowski, S. Paranjape, L. Ren, M. Skowronek, M. Spradlin, A. Volovich, **H.-C. Weng** and Y.-Q. Zhang, *de Sitter Wavefunction from Quadrangular Polylogarithms: Loop Graphs*

Seminar Talks

1. QFT Group Seminar, Max Planck Institute for Physics Dec 2025
Seminar talk: *Split-Helicity Tree Amplitudes and Flag Cluster Algebras*

Short Talks and Posters

- | | |
|---|----------|
| 1. Prospects in Theoretical Physics 2026 Summer School
Short talk: <i>De Sitter Wavefunction from Quadrangular Polylogarithms: Chains</i> | Jul 2026 |
| 2. Amplitudes 2026 Conference
Short talk: <i>De Sitter Wavefunction from Quadrangular Polylogarithms: Chains</i> | Jun 2026 |
| 3. Symbology@15 Conference
Short talk: <i>Split-Helicity Tree Amplitudes and Flag Cluster Algebras</i> | Dec 2025 |
| 4. Michigan Interdisciplinary Meeting on Amplitudes: Bridges between Physics & Mathematics
Short talk: <i>The cluster adjacency of split-helicity Yang-Mills</i> | Oct 2025 |
| 5. Amplitudes 2025 Conference
Poster: <i>Symbol Alphabets in QCD and Flag Cluster Algebras</i> | Jun 2025 |
| 6. Amplitudes 2024 Summer School
Poster: <i>Unitarity of bespoke amplitudes</i> | Jun 2024 |
| 7. Amplitudes 2024 Conference
Short talk: <i>Unitarity of bespoke amplitudes</i> | Jun 2024 |

Selected Conferences and Workshops Participation

- | | |
|---|--------------|
| 1. Amplitudes 2026 School | Jul 2026 |
| 2. Lancefest Workshop | Jun 2026 |
| 3. Max Planck-IAS-NTU Center Kick-off Conference | Mar 2026 |
| 4. From Analyticity to Phenomenology Workshop | Jan 2026 |
| 5. Symbology@15 Conference | Dec 2025 |
| 6. Amplitudes Afterparty in Taiwan | Jun 2025 |
| 7. Amplitudes 2025 Summer School | Jun 2025 |
| 8. Summer School in Total Positivity and Quantum Field Theory | Jun 2025 |
| 9. QCD Meets Gravity 2024 | Dec 2024 |
| 10. Amplituhedra, Cluster Algebras, and Positive Geometry Conference | May 2024 |
| 11. Mathematical Aspects of Scattering Amplitudes Program | Apr-May 2024 |
| 12. Kickoff Workshop for Simons Collaboration in Celestial Holography | Oct 2023 |

Academic Honors

- | | |
|------------------------------------|-------------|
| 1. Physics Dissertation Fellowship | 2026 - 2027 |
|------------------------------------|-------------|

Academic Appointments

- | | |
|--|------------------------------------|
| 1. Research Assistant, PIs: Prof. Spradlin & Prof. Volovich | Summer 2023; Summer 2025 - Present |
| 2. Teaching Assistant, PHYS2060 - Quantum Mechanics | Spring 2025 |
| 3. Teaching Assistant, PHYS0040 - Basic Physics | Fall 2024 |
| 4. Teaching Assistant, PHYS2060 - Quantum Mechanics | Spring 2024 |
| 5. Teaching Assistant, PHYS1530 - Thermodynamics and Statistical Mechanics | Fall 2023 |